A guided macro example

1 Give notation a name

The set of real numbers is $\mathbb{R}$. Inside mathematics the same command works: $x \in \mathbb{R}$.

Let $U(c, s) = \log c + s$ with $c > 0$. The first and second arguments of the partial-derivative macro are the function and its variable:

$$\frac{\partial U}{\partial c} = \frac{1}{c}, \qquad \frac{\partial U}{\partial s} = 1.$$

2 Use an optional label

Note: The first argument has a default value.

Input: An explicit label replaces that default.

3 Read a star

Let x and y be vectors in the same space. The unstarred norm has ordinary delimiters:

$$\|x + y\|.$$

The starred call sizes delimiters to the taller argument:

$$\left\| \frac{x + y}{2} \right\|.$$

The star state is argument one; the norm's content is argument two.

4 Map a comma list

The public command below passes a comma list to an internal function:

- Input
- Calculation
- Output

Inside the enclosing function definition, the current list item uses a doubled parameter marker. Removing one input word should remove one list item without changing the function.